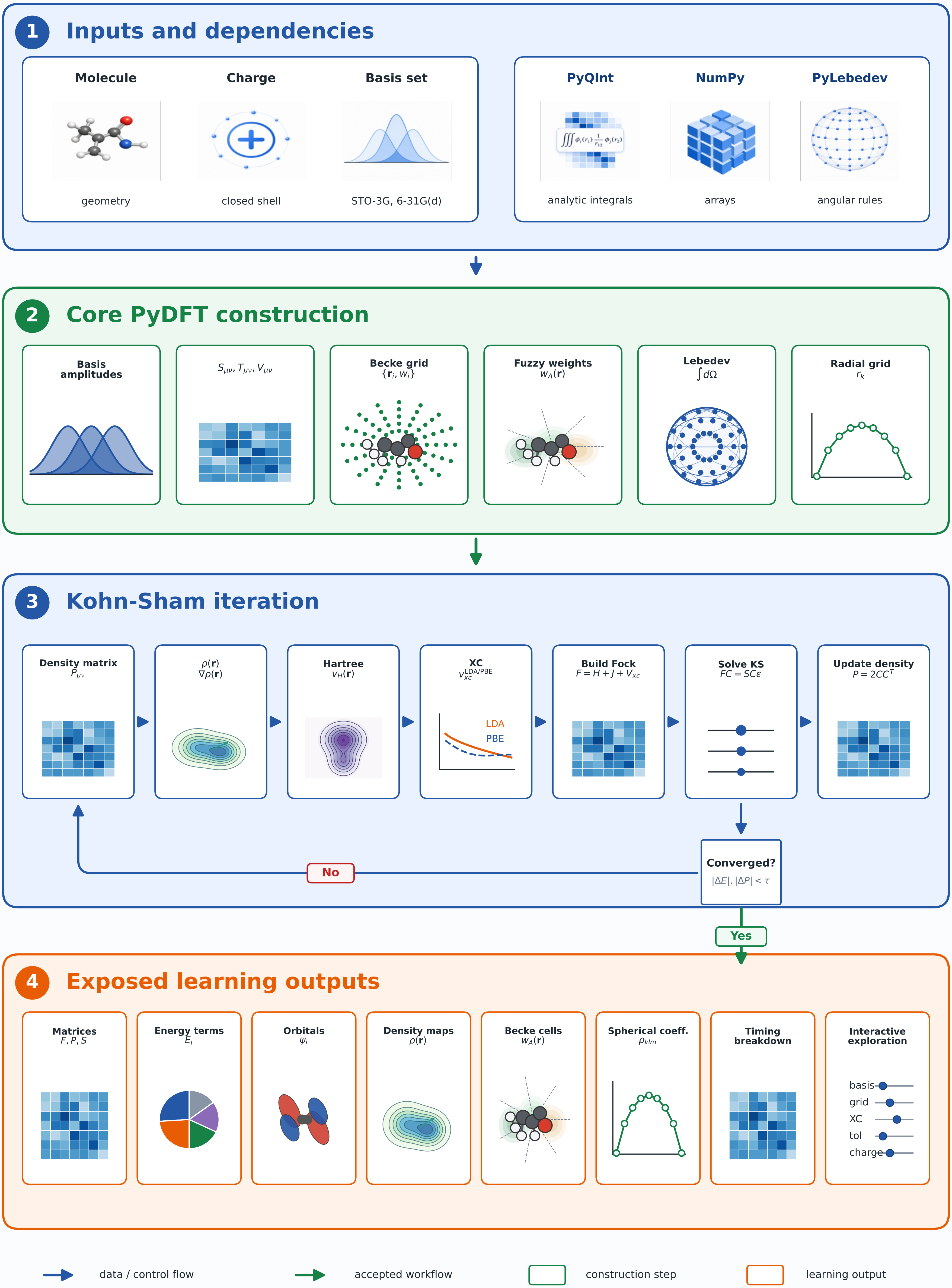
1 Inputs and dependencies
Molecule
geometry
Charge
closed shell
Basis set
STO-3G, 6-31G(d)
PyQInt
analytic integrals
NumPy
arrays
PyLebedev
angular rules
2 Core PyDFT construction
Basis amplitudes
$S_{\mu\nu}, T_{\mu\nu}, V_{\mu\nu}$
Becke grid $\{\mathbf{r}_i, w_i\}$
Fuzzy weights $w_A(\mathbf{r})$
Lebedev $\int d\Omega$
Radial grid r_k
3 Kohn-Sham iteration
Density matrix $P_{\mu\nu}$
$\rho(\mathbf{r})$ $\nabla\rho(\mathbf{r})$
Hartree $v_H(\mathbf{r})$
XC $v_{xc}^{\mathrm{LDA/PBE}}$
LDA
PBE
Build Fock $F = H + J + V_{xc}$
Solve KS $FC = SC\varepsilon$
Update density $P = 2CC^T$
Converged? $|\Delta E|, |\Delta P| < \tau$
No
Yes
4 Exposed learning outputs
Matrices F, P, S
Energy terms E_i
Orbitals ψ_i
Density maps $\rho(\mathbf{r})$
Becke cells $w_A(\mathbf{r})$
Spherical coeff. ρ_{klm}
Timing breakdown
Interactive exploration
basis
grid
XC
tol
charge
data / control flow
accepted workflow
construction step
learning output